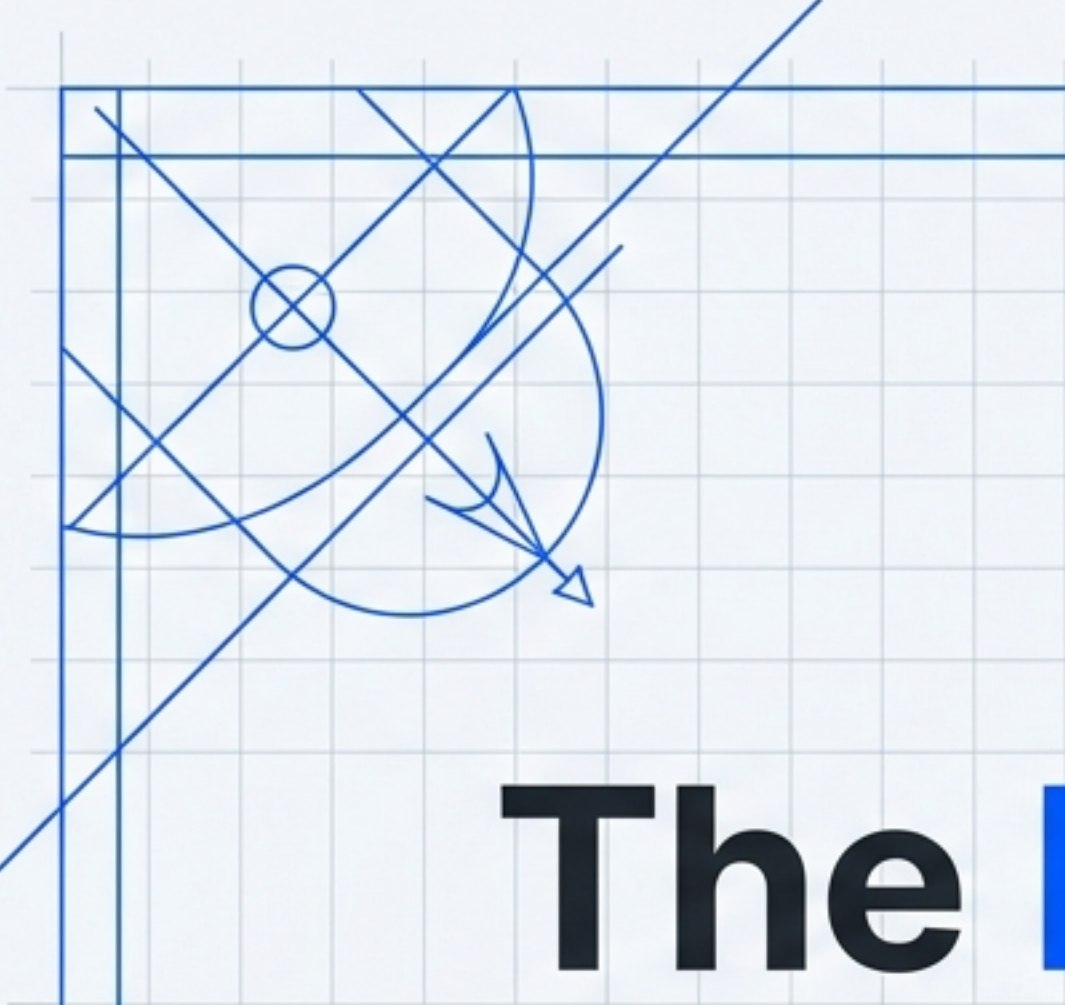




The **Blueprint** of Data

A Foundational Guide to **Statistics** for Data Scientists

Our Blueprint for Today

1. Defining the Discipline

2. Classifying the Materials

3. Applying the Concepts

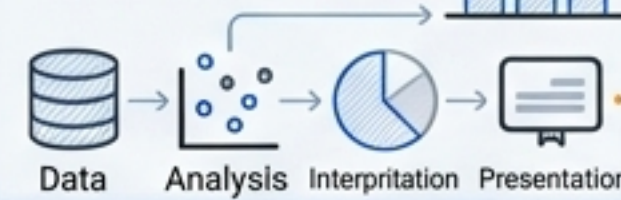
4. Visualizing the Structure

5. Understanding the Scope

1

WHAT IS STATISTICS?

the science of collecting, analyzing, interpreting and presenting data



2

TYPES OF DATA

NUMERICAL

123

DISCRETE



NOMINAL

CATEGORICAL

CONTINUOUS

Low High

ORDINAL

3

REPRESENTATION OF DATA



4



5

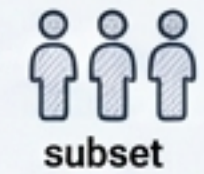
POPULATION vs SAMPLE

POPULATION



entire group
Example:
All students

SAMPLE



subset
of population
Example:
500 students

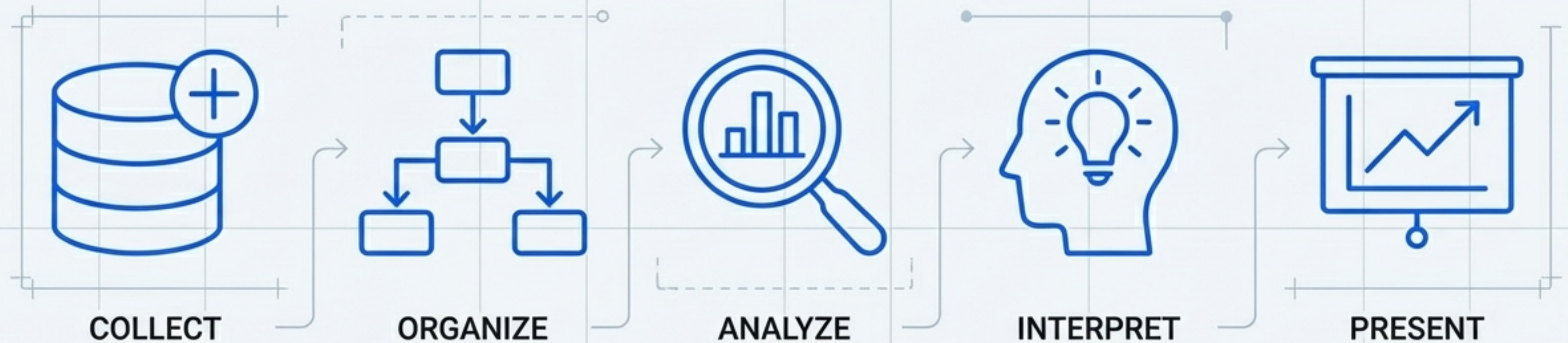
DATA IN STATISTICS

Applications of statistics

- Healthcare
- Finance
- Marketing
- Sports

Statistics is the Science of Data

A field of mathematics considered the science of **collecting, organizing, analyzing, interpreting, and presenting** data.



Why It's the Core of Data Science



→ **Data Analysis:** To understand trends and patterns.



→ **Predictive Modeling:** To build machine learning models.



→ **Decision Making:** To optimize business strategies with data-driven insights.



→ **Data Cleaning:** To detect outliers, missing values, and anomalies.



→ **A/B Testing:** To evaluate and measure the impact of changes.

Data is the Raw Material for Insight

Data refers to **raw facts, figures, or information** collected for analysis.

Its Applications are Universal



Healthcare

Disease prediction,
patient analysis.



Finance

Risk assessment,
fraud detection.



Marketing

Customer behavior analysis,
campaign performance.



Sports

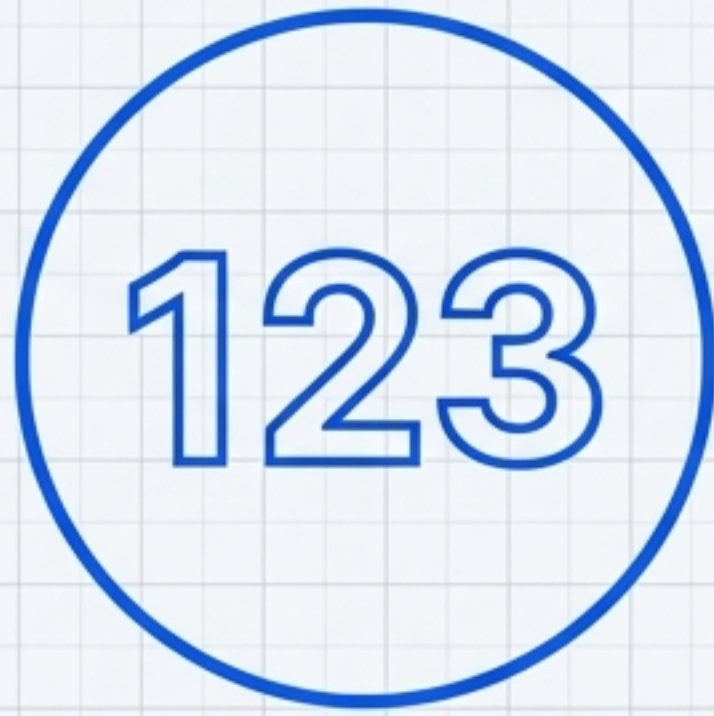
Performance tracking,
game strategy.



Government

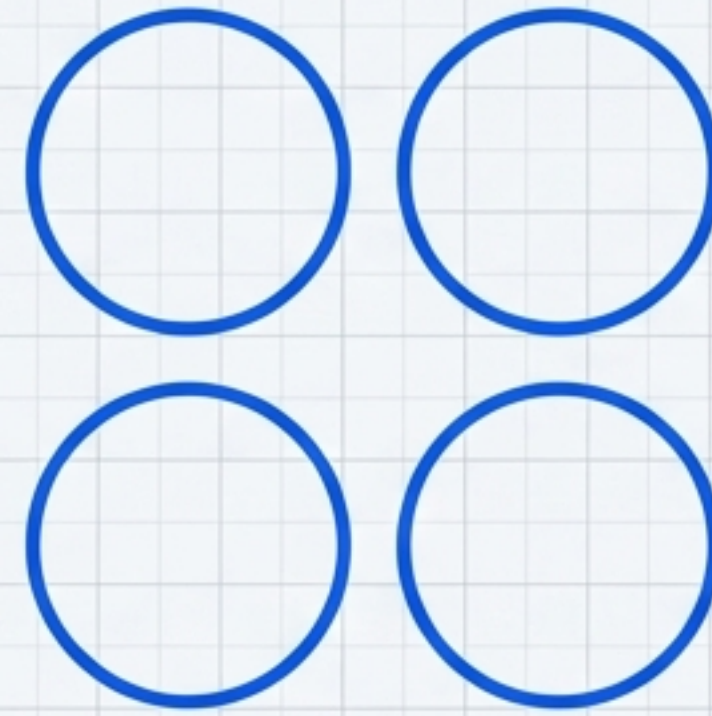
Census data, unemployment
rates, economic indicators.

The Two Fundamental Types of Data



Numerical (Quantitative)

Represents measurable quantities. Data that can be counted or measured.

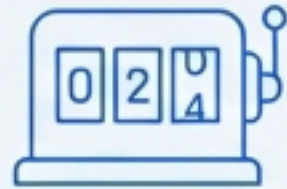


Categorical (Qualitative)

Represents descriptive labels or groups. Data that can be observed but not measured.

Deconstructing Numerical Data: Discrete vs. Continuous

Countable Things



Consists of whole, countable numbers.
There are finite or countably infinite values.

- Number of children
- Quantity of products in a cart
- Number of students in a seminar (e.g., 35)

Measurable Things



Can take any value within a given range.
Often includes decimals.

- Height (e.g., 6.2 ft)
- Weight
- Temperature
- Price

PRO-TIP FOR CODERS

Think of **Discrete** data as an integer and **Continuous** data as a float.

Visualizing the Difference

Number of Students in a Hall (Capacity 100)



Height in Feet



Deconstructing Categorical Data: Nominal vs. Ordinal

Categories with No Order



Data that represents categories without any intrinsic ranking or order.

"***NO***minal" = "**No** Specific Order.

- Gender (Male, Female)
- Colors (Red, Blue, Green)
- Country (USA, India)

Categories with an Order



Data that represents categories with a meaningful, predefined order or rank.

"***ORD***inal" = "**Ord***ered Categories.

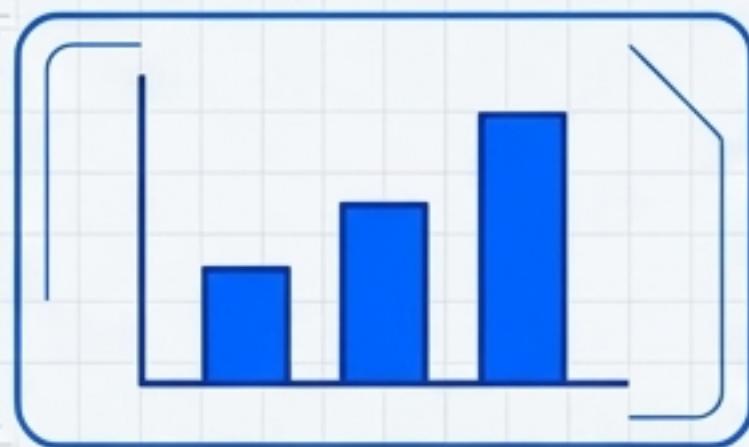
- Satisfaction Level (Low, Medium, High)
- Grades (A, B, C)
- Months of the year

The Complete Data Classification Matrix

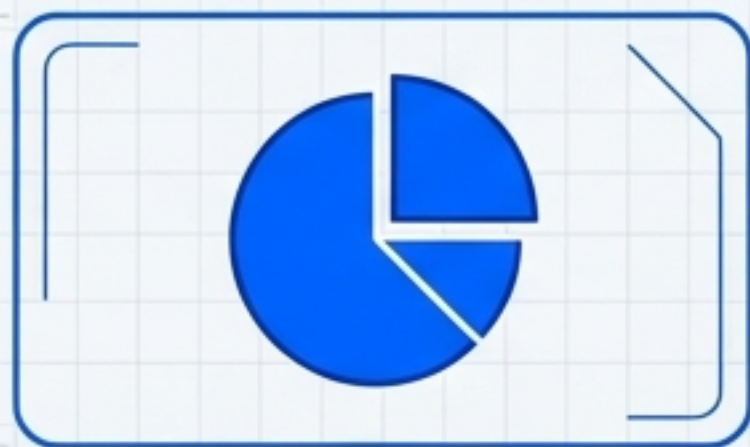
TYPE	CATEGORY	DESCRIPTION	EXAMPLES
Numerical (Quantitative)			
➤ Discrete		Countable numbers	Number of children, Product quantity
➤ Continuous		Measurable quantities with decimals	Height, Weight, Temperature
Categorical (Qualitative)			
➤ Nominal		Categories with no spec	Gender (Male/Female), Country
➤ Ordinal		Categories with a meani	Ratings (Low, Medium, High)

From Classification to Representation

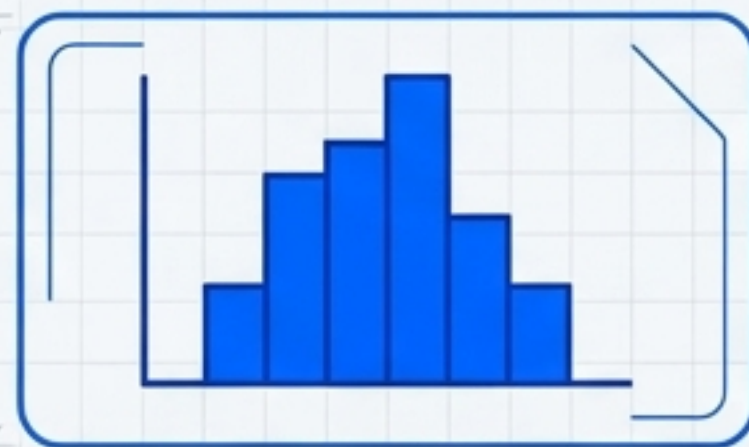
Once we know the type of our data, we can choose the right visual tool to uncover its story. This is the art of graphical representation.



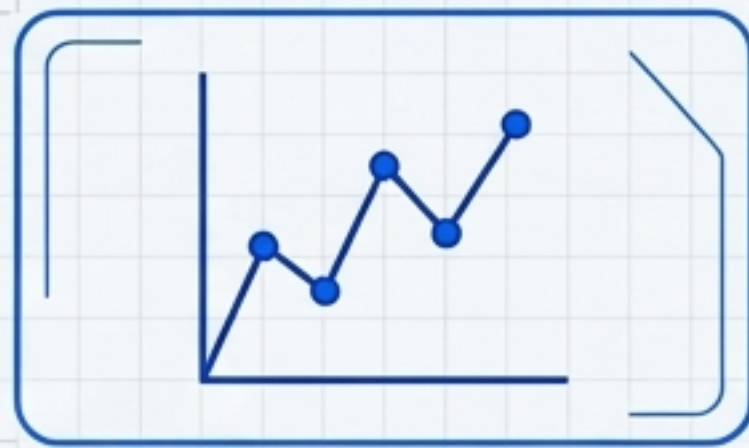
BAR CHART



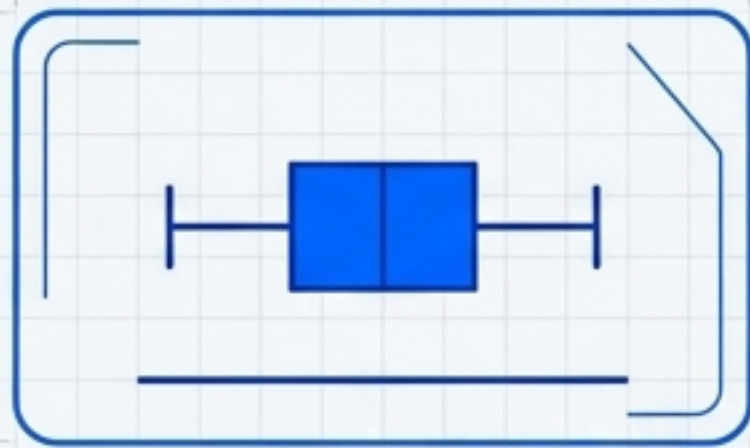
PIE CHART



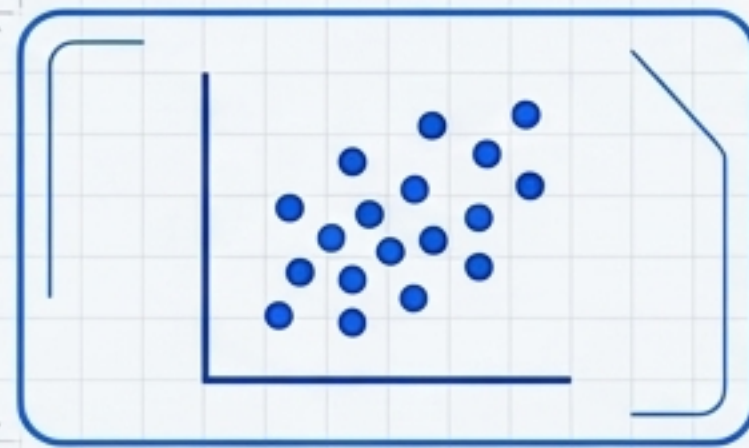
HISTOGRAM



LINE GRAPH

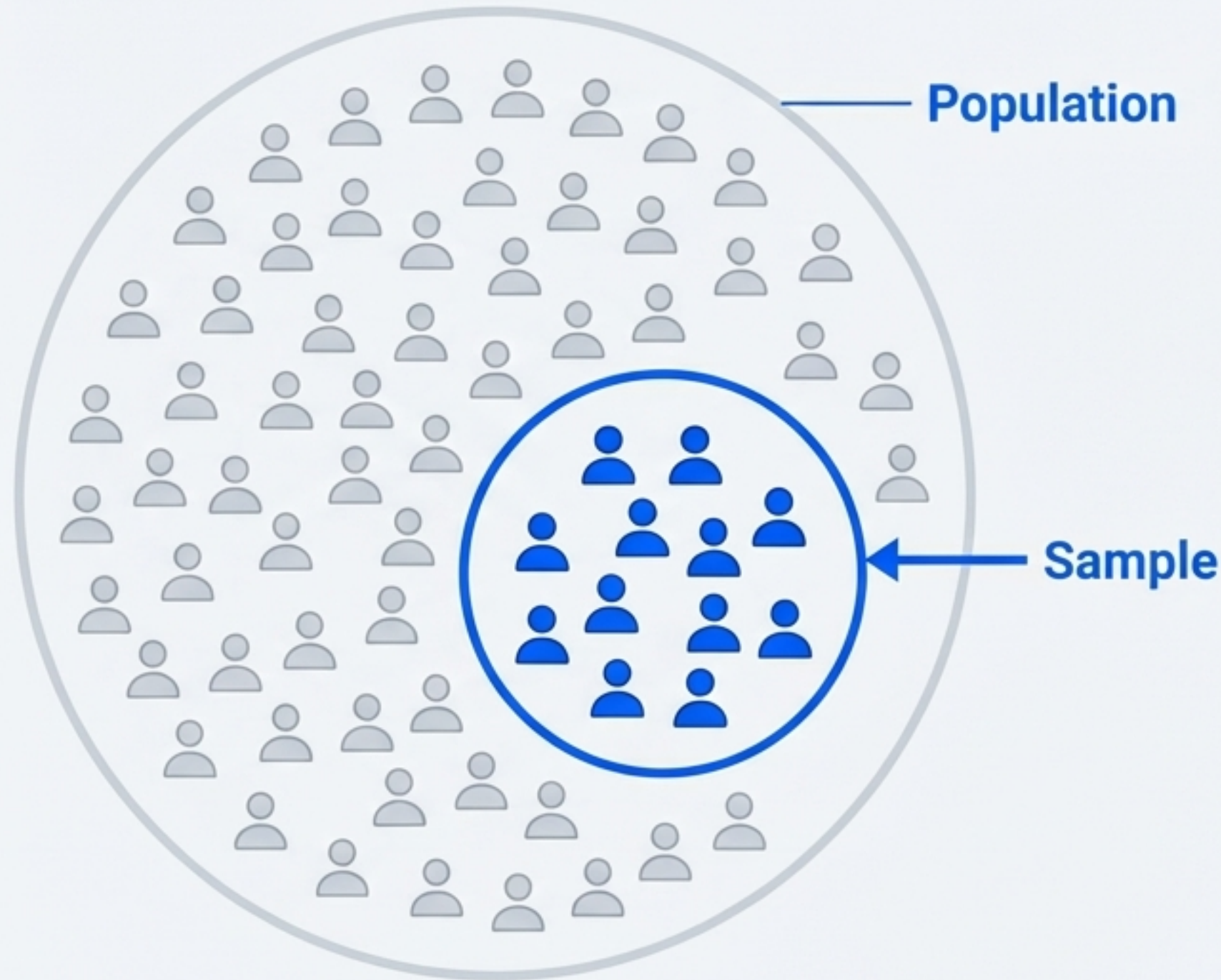


BOX PLOT



SCATTER PLOT

Scope of Study: Population vs. Sample



Population

Population

The entire group you want to study.

* *Analogy: The complete project blueprint.*

* *Example: All students in India.*

Sample

Sample

A smaller, representative part selected from the population.

* *Analogy: A detailed scale model.*

* *Example: 500 students selected from Gujarat.*

Why Work with a Sample?



Efficiency

Faster and cheaper than studying an entire population.



Practicality

Often, studying the entire population is impossible.



Accuracy

A properly chosen sample can yield accurate results about the population.

The Power of Inference

We use a process called **statistical inference** to make educated conclusions about the entire population based on the findings from our sample.

Your Completed Blueprint

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Data



Analysis



Interpretation



Presentation

TYPES OF DATA

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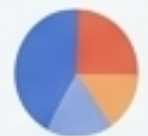


ORDINAL

REPRESENTATION OF DATA



BAR CHART



PIE CHART



HISTOGRAM



BOX PLOT



SCATTER PLOT



SCATTER PLOT

POPULATION vs SAMPLE



entire group

Example:

All students



subset

of population

Example:

500 students

DATA IN STATISTICS

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—	—
—	—
—	—

Applications of statistics

- Healthcare
- Finance
- Marketing
- Sports

“Knowing these terms is what separates a skilled practitioner from a professional expert. It’s the difference between a mechanic who can fix a car and an engineer who knows *why* the fix works. You are now building the foundation for professional expertise.”